

HEREDITARY BREAST AND OVARIAN CANCER PANEL (12 GENES) — FINAL REPORT

PATIENT BERGSTROM, ANITA L. MRN MRN-5519082 52 y / Female	SPECIMEN Lab accession ATLAS-26-G-06224 Client WCI-26-S-04115 Peripheral blood, EDTA Drawn 2026-05-01	ORDERING Westhaven Cancer Institute T. Okafor, MD Received 2026-05-04 Reported 2026-05-13
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Note: this report was received after the primary surgical pathology sign-out (2026-04-29) and constitutes an ancillary germline component of the integrated diagnostic episode. The integrated report should be amended to incorporate this result.

RESULT SUMMARY

NEGATIVE — No pathogenic or likely pathogenic variants identified in any of the 12 genes on this panel. No reportable variants of uncertain significance (VUS) in BRCA1, BRCA2, PALB2, ATM, CHEK2, or TP53.

GENES TESTED (NEGATIVE FOR P / LP VARIANTS)

Gene	Coverage (mean)	Result
BRCA1	245x	No reportable variant
BRCA2	238x	No reportable variant
PALB2	227x	No reportable variant
ATM	210x	No reportable variant
CHEK2	232x	No reportable variant
TP53	260x	No reportable variant
CDH1	215x	No reportable variant
STK11	220x	No reportable variant
PTEN	243x	No reportable variant
RAD51C	229x	No reportable variant
RAD51D	224x	No reportable variant
NF1	201x	No reportable variant

INTERPRETATION

Germline testing across 12 hereditary breast and ovarian cancer genes returns **NEGATIVE** for pathogenic and likely pathogenic variants. Notably, BRCA1, BRCA2, PALB2, ATM, CHEK2, and TP53 are fully covered (mean coverage > 200x; minimum 50x per exon). No reportable VUS were identified. Given the patient's positive family history (mother breast Ca at 68, maternal aunt ovarian Ca at 62), this result does not establish a familial hereditary predisposition syndrome but does **NOT** exclude a low-penetrance non-coding variant, an undetected large rearrangement outside panel coverage, or a syndrome conferred by a gene not on this panel. Continued risk-adapted surveillance of first-degree relatives per current institutional guidelines is reasonable.

METHODOLOGY & LIMITATIONS

DNA extracted from peripheral blood lymphocytes; library prepared with hybrid capture; sequenced on a high-throughput platform with mean target coverage > 200x per gene. Variant calls assessed per ACMG/AMP 2015 + ClinGen guidance; variants classified as pathogenic, likely pathogenic, VUS, likely benign, or benign. Variants of uncertain significance, likely benign, and benign are not reported here unless directly affecting the requested decision. Large rearrangements assessed by MLPA for BRCA1, BRCA2, and PALB2. Mosaicism below 10% may not be detected. Atlas Hereditary Cancer Genetics is a CLIA-certified, CAP-accredited laboratory; this is a laboratory-developed test.